Project Documentation

1. Project Title

Student Performance Analytics System

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2. Objective

The primary objective of this project is to build a Python-based Student Performance Analytics System. It solves the problem of manually calculating student grades and performance metrics by automating the reading, processing, and analysis of student records to output meaningful results such as pass/fail statistics, subject averages, and top performers.

3. Technologies Used

- **Python:** Main programming language utilized for core logic.
- **Pandas:** Used for loading the CSV, organizing the data into a structured DataFrame, and seamless data manipulation.
- **NumPy:** Used for numerical operations like calculating total sums and means across multiple subjects efficiently.

4. Dataset Description

The dataset, saved as `students.csv`, contains 15 detailed student records. Important columns include `Student_ID`, `Name`, `Department`, subject marks (`Math_Marks`, `Science_Marks`, `English_Marks`), and `Attendance_Pct`. Realistic integer and percentage values were utilized for all fields to accurately simulate a classroom environment.

5. Implementation Workflow

- **Data Loading:** Read `students.csv` using `pd.read_csv()` to initialize the Pandas DataFrame.
- **Data Processing:** Utilized `np.sum()` and `np.mean()` along axis 1 to calculate total marks and average marks for each student.
- **Functions & Loops:** Created reusable functions `assign_grade` and `check_pass_fail` in a separate module. A `for` loop utilizing `df.iterrows()` was implemented to apply these functions to every record based on their specific average and attendance criteria.
- **Data Analysis:** Used NumPy's `np.max()`, `np.min()`, and `np.mean()` to extract overall class metrics. Filtered the DataFrame to count passing students and locate the top performers.

6. Key Features

- Automated total and average calculations using NumPy arrays.
- Modular grading logic and strict pass/fail evaluation via custom functions.
- Detailed class summary including the highest, lowest, and overall class average scores.

7. Output

Below is the terminal output representing the successful execution of `main.py`:

```
--- Student Performance Analytics System --- Total student records loaded: 15 --- Performance Analysis --- Class Average: 71.4 Highest Average: 95.0 Lowest Average: 37.67 Pass/Fail Statistics: 11 Passed, 4 Failed --- Subject-wise Performance (Average Marks) --- Math: 70.47 Science: 72.8 English: 70.93 --- Top Performing Student(s) --- Name: Krishna Iyer | Department: Computer Science | Marks: 285 --- Processed Student Data (Top 5 Rows) --- Name Total_Marks Average_Marks Grade Status 0 Aarav Sharma 263 87.67 A Pass 1 Vivaan Singh 137 45.67 D (Fail) Fail 2 Aditya Patel 235 78.33 B Pass 3 Vihaan Kumar 276 92.00 A Pass 4 Arjun Rao 113 37.67 D (Fail) Fail
```

8. Final Outcome

The project successfully loaded the raw student dataset, processed it accurately without errors, and produced a clear terminal report displaying class averages, subject performance, pass/fail counts, and identified the top performer. The dataset is properly structured, and the final output is understandable without needing to inspect the underlying Python code.

9. Challenges & Learning

Challenge: Calculating row-wise sums across specific subject columns without hardcoding values or causing Pandas index errors.

Solution & Learning: Used Pandas column selection combined with NumPy's `np.sum(axis=1)`. This provided valuable practical experience in integrating Pandas DataFrames with NumPy numerical calculations, and utilizing iteration safely within data analysis workflows.

10. GitHub Repository

Repository Link: https://github.com/pythonlearner793/student-performance-analytics.git